

N322 Model – V1.74 (J, K, T) | V1.75 (Pt100)

Table of Registers for Serial Communication

1 COMMUNICATION SERIAL

1.1 RS485 INTERFACE

- Compatible line signals with RS485 standard.
- 2-wire connection from master to up to 31 slaves in a multidrop bus. It is possible to address 247 nodes with multiple output converters.
- Maximum communication distance: 1000 meters.
- The RS485 signals are:

D1	D	D+	B	Bidirectional data line.
D0	$\overline{D}$	D-	A	Inverted bidirectional data line.
C				Communication common. Interconnect between all network devices for protection.
GND				

Table 1

1.2 GENERAL CHARACTERISTICS

- The serial interface is not isolated from input circuitry.
- The serial interface is isolated from input circuitry, except in 24 V powered model.
- Baud rate: 9600 bps.
- Data Bits: 8.
- Parity: None.
- Stop Bits: 1.

1.3 COMMUNICATION PROTOCOL

The device is compatible with Modbus RTU slave protocol, available for most SCADA software on the market.

The available Modbus commands are:

03	Read Holding Register
06	Preset Single Register

Command 02 (Read Holding Register) can read up to 4 registers in the same request.

2 CONTROLLER CONFIGURATION

Controllers with RS485 serial interface display the **Addr** parameter on their programming level. In this parameter, user can set a **communication address** for each element on the network. The address must be set between 1 and 247.

Addr	Device communication address. Each device must have an exclusive address.
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3 TABLE OF REGISTERS

The registers are the internal parameters of the controller.

Each parameter is a 16-bit word with negative values represented as 2's complement.

HOLDING REGISTERS	PARAMETER	DESCRIPTION
0000	SP1	Read: OUTPUT1 Setpoint (Control output). Write: OUTPUT1 Setpoint (Control output). Range: From SP1 to the value specified in SPH .

HOLDING REGISTERS	PARAMETER	DESCRIPTION
0001	PV	Read: Value of the measured temperature. Write: Not allowed. Range: According to the sensor type used.
0002	IHM Status1	Read: IHM Status. Write: Not allowed. Value format: Bit 0 – OUT1 indicator. Bit 1 – OUT2 indicator. Bit 10 – Decimal point. Bit 12 – Signo.
0003	Control Status1	Read: Measurement and OUTPUT1 status. Write: Not allowed. Value format: Bit 0 – Underflow measurement. Bit 1 – Overflow measurement. Bit 8 – OUTPUT1 status. Bit 13 – Controller in defrost.
0004	Value of the parameter being displayed	Read: Value of the parameter being presented on the display. Write: Not allowed. Maximum range: -199 to 999. The actual range depends on the parameter being displayed.
0005	Version No. Screen	Read: Software version implemented in the controller and current screen number. Write: Not allowed. Value format: XYYh, where: XX → Firmware version. YY → Screen number.
0006	Serial number (High)	Read: First 3 digits of the controller serial number. Write: Not allowed. Value format: XXXh.
0007	Serial number (Low)	Read: Last 3 digits of the controller serial number. Write: Not allowed. Value format: XXXh.
0008	Hysteresis 1	Read: OUTPUT1 hysteresis. Write: OUTPUT1 hysteresis. Range: 0.1 to 50.0.
0009	Control Status 2	Read: OUTPUT2 Status. Write: Not allowed. Value format: Bit 0 – Status de OUTPUT2.
0010	SP2	Read: OUTPUT2 Setpoint. Write: OUTPUT2 Setpoint. Range: From SP1 to the value specified in SPH .
0011	Hysteresis 2	Read: OUTPUT2 hysteresis. Write: OUTPUT2 hysteresis. Range: 0.1 to 50.0.
0012	Offset	Read: Offset value. Write: Offset value.

HOLDING REGISTERS	PARAMETER	DESCRIPTION
		Range: -10.0 to 10.0.

Table 2

Note: The SP, PV and Hysteresis values are always multiplied by 10 to account for the decimal point.

4 EXCEPTION RESPONSES: ERROR CONDITIONS

Upon receiving a command, the device performs a CRC check on the received data block. If a CRC error is detected, no response will be sent to the master.

For commands received without errors, a consistency check is performed on the command and the requested registers; if they are invalid, an exception response is returned with the corresponding error code. In exception responses, the Modbus function code in the reply is the request's function code plus 0x80.

If a written command includes a value outside the permitted range, no value is written to that parameter and error code 03 is returned.

Broadcast reading commands are ignored by the controller. It is only possible to write in Broadcast mode.

ERROR CODE	ERROR DESCRIPTION
01	Invalid command.
02	Invalid register number or out of range.
03	Invalid register quantity or out of range.

Table 3

5 ELECTRICAL CONNECTIONS

5.1 PHYSICAL ENVIRONMENT CHARACTERISTICS

- Shielded twisted-pair cable, 3x 24 AWG, and grounded mesh at both ends.
- Capacitance of 16 pF per 30 m.
- Impedance of 100 Ω.
- Maximum length of 1000 m.

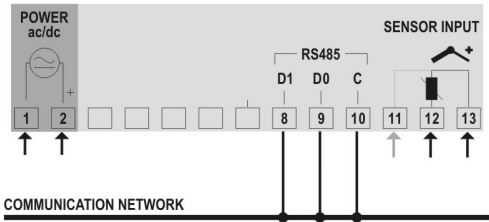


Figure 1